Final Exam Review

ECON 308 – International Economic Relations

Fall 2025

Problem 1 — Ricardian Model

Two countries produce wheat (W) and cloth (C). Unit labor requirements:

Country	a_{LW}	a_{LC}
Home	4	2
Foreign	5	1

Home has 200 workers; Foreign has 150.

(1) Determine each country's autarky relative price of wheat, P_W/P_C.

(2) Identify who has comparative advantage in each good.

(3) Suppose the world relative price is $P_W/P_C = 1.5$.

- Which good does each country export?
- Compute each country's production bundle under free trade.

(4) Explain why both countries gain from trade even if one is absolutely less productive.

Problem 2 — Specific Factors Model

Home produces food (F) and manufactures (M). Labor is mobile; capital is specific to M; land is specific to F.

$$Q_M = 5K^{1/2}L_M^{1/2}, \qquad Q_F = 4T^{1/2}L_F^{1/2}.$$

Total labor: $L = 200$. Capital: $K = 100$. Land: $T = 100$.

(1) Derive the marginal product of labor in each sector.

(2) Set up the equilibrium condition for labor allocation.

(3) Suppose the relative price of manufactures rises.

- What happens to wages?
- What happens to the return to capital and the return to land?

(4) Explain who gains and who loses from this price change.

Problem 3 — Heckscher–Ohlin Model

Two goods: machinery (M) is capital-intensive; textiles (T) is labor-intensive.

Country	K	L
Home	900	400
Foreign	300	600

(1) Identify which country is capital-abundant and which is labor-abundant.

(2) According to Heckscher–Ohlin, who exports which good?

(3) Using Stolper–Samuelson, explain who gains and who loses from opening to trade in each country.

(4) Using Rybczynski, describe how an increase in Home labor affects output of M and T.

Problem 4 — Standard Trade Model

A country produces apples (A) and electronics (E):

$$A^2 + 4E^2 = 1600.$$

World price ratio:

$$P_E/P_A = 2.$$

Preferences: $E = 0.25A$.

(1) Find the production point that maximizes the value of output.

(2) Determine the consumption bundle.

(3) Calculate exports and imports.

(4) Suppose P_E/P_A rises.

- Explain how production and consumption adjust.
- Does welfare rise or fall?

(5) Define export-biased growth and explain its effect on the terms of trade.

Problem 5 — External Economies of Scale and Location

Two countries A and B can produce semiconductors. There are external economies of scale, so average cost falls with industry size. A currently hosts the industry; technologies are identical.

(1) Explain why A may remain the global semiconductor producer even though B has the same technology.

(2) Describe conditions under which B could successfully enter the industry.

(3) Suppose B subsidizes semiconductor production.

- Explain how this may shift global production to B.
- Discuss welfare implications for A, for B, and globally.

Problem 6 — Trade Policy (Tariffs)

Demand and supply for steel in Home:

$$Q_D = 800 - 2P, \qquad Q_S = P - 40.$$

World price: $P_W = 200$. Foreign export supply:

$$P^* = 150 + 0.5Q^*.$$

(1) Find the free-trade price and quantity of imports.

(2) Home imposes a tariff of $t = 30$.

- Find the new world price.
- Find the new domestic price.
- Compute imports.

(3) Compute:

- Tariff revenue
- Terms-of-trade gain
- Deadweight loss

(4) Determine whether national welfare rises or falls.